

Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition 1: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

A URL (Uniform Resource Locator) is a unique address used to locate and access resources on the internet.

- Identifies the location of a resource online.
- Specifies the protocol (e.g., HTTP or HTTPS).
- Includes the domain name or IP address.
- May contain a path to a specific resource.
- Used to access web pages, files, images, and other online resources.

Parts of URL

A URL is composed of multiple parts, each defining how and where a resource can be accessed.

- **Scheme:** Defines the protocol used to access the resource (e.g., http, https, ftp).
- **Subdomain:** Specifies a subdivision of the main domain (e.g., www).
- **Domain:** Identifies the main website name (e.g., example).
- **Top-Level Domain (TLD):** Indicates the domain extension (e.g., .com, .org).
- **Port Number:** Specifies the server port (default: 80 for HTTP, 443 for HTTPS).
- **Path:** Points to the resource location on the server.
- **Query String Separator:** The ? symbol that starts query parameters.
- **Query String / Parameters:** Passes data as key–value pairs using &.
- **Fragment:** Refers to a specific section within the resource using #.

Source: <https://www.geeksforgeeks.org/computer-networks/components-of-a-url/>

2. How does HTML file process in web browser?

When you visit a website, a specific process happens called the request-response cycle. This is the heart of how the web works.

Step 1: You Enter a URL

You type a web address (URL) like www.example.com into your browser's address bar and press enter. This tells your browser where you want to go.

The URL is like a street address. It tells the browser exactly which server to contact and which file to request.

Step 2: Browser Sends a Request

Your browser sends a request over the internet to the server that hosts the website. This request travels through your internet connection, your internet service provider, and various network devices.

The request is asking the server: "Can you send me the HTML file for this webpage?"

Step 3: Server Processes the Request

The server receives your browser's request and looks for the requested HTML file. If the file exists, the server prepares to send it back.

If the file doesn't exist, the server will send back an error message (like the famous "404 Not Found" error).

Step 4: Server Sends a Response

The server sends the HTML file back to your browser over the internet. This response includes the HTML code and other information about the file.

This happens incredibly fast, usually in fractions of a second, depending on your internet speed and the server's location.

Step 5: Browser Displays the Page

Your browser receives the HTML file and reads the code inside. It then translates that code into the visual webpage you see on your screen.

The browser follows the instructions in the HTML to display headings, paragraphs, images, and other elements in the correct order and style.

Source: <https://unwiredlearning.com/blog/html-basics-browsers-servers>

3. How will page navigation flow managed among all HTML pages?

Sitemaps are what you might expect — maps of all the different pages (or the main sections if you are talking about really huge sites) of your site. They allow your site's visitors to get a glimpse of the overall structure of your site, and go anywhere they need to fast — even if the page they need is deep within your page hierarchy.

Both sitemaps and site searches are a great way of offering visitors a fallback option when they are lost or to offer quick access for those who are in a hurry.

From an HTML point of view they could either be one massive nested list full of links or — in the case of very large sites — section headings with nested links of section-specific hierarchies or even search forms for each of the sections.

Source:

https://www.w3.org/wiki/Creating_multiple_pages_with_navigation_menus#Sitemaps

4. How will GitHub commits be maintained after each practical?

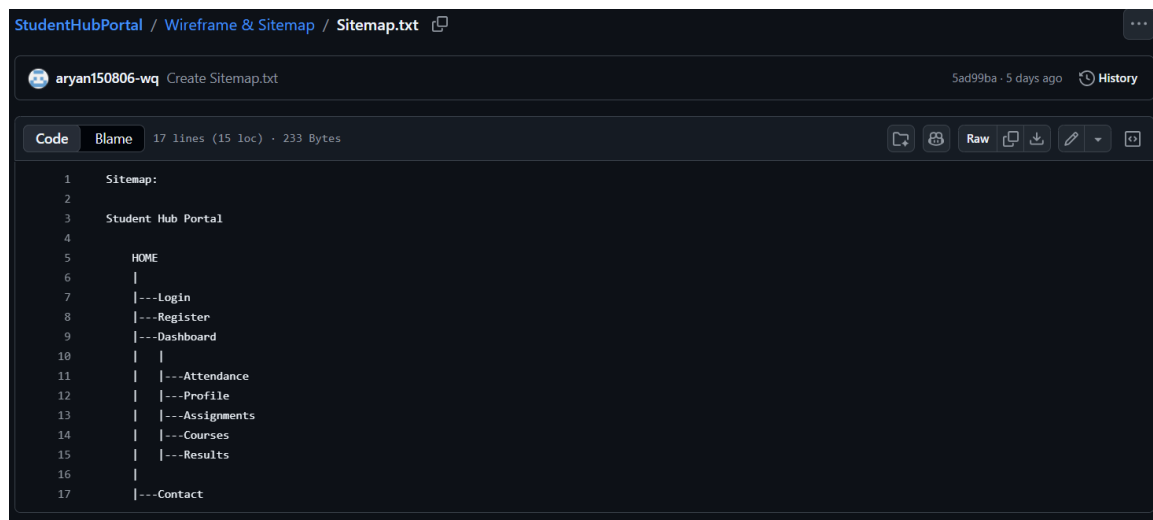
Answer: GitHub commits should be maintained by making a commit after completing each practical. Every commit should include only the changes made for that particular practical and have a clear, meaningful commit message describing what was done. This creates a record of the work completed over time and makes it easy to track progress.

Example Commit Messages

- Added Practical 1 - HTML Basics
- Completed Practical 2 - Lists and Tables
- Updated Practical 3 - Forms
- Fixed errors in Practical 4

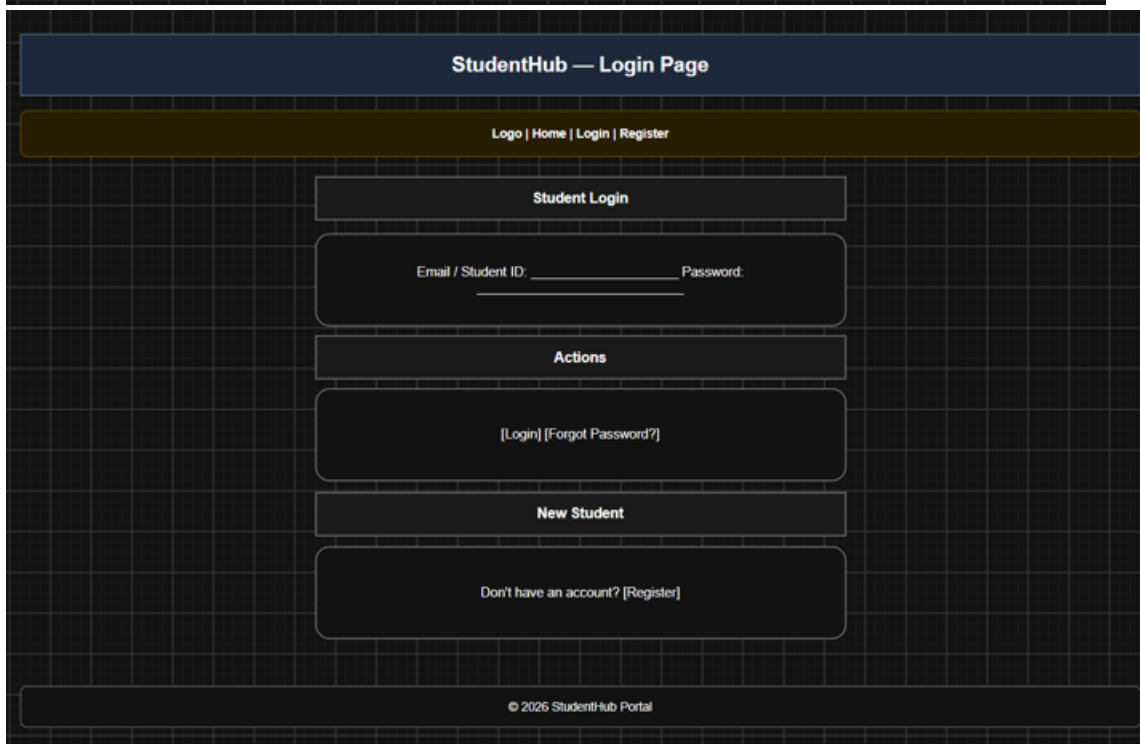
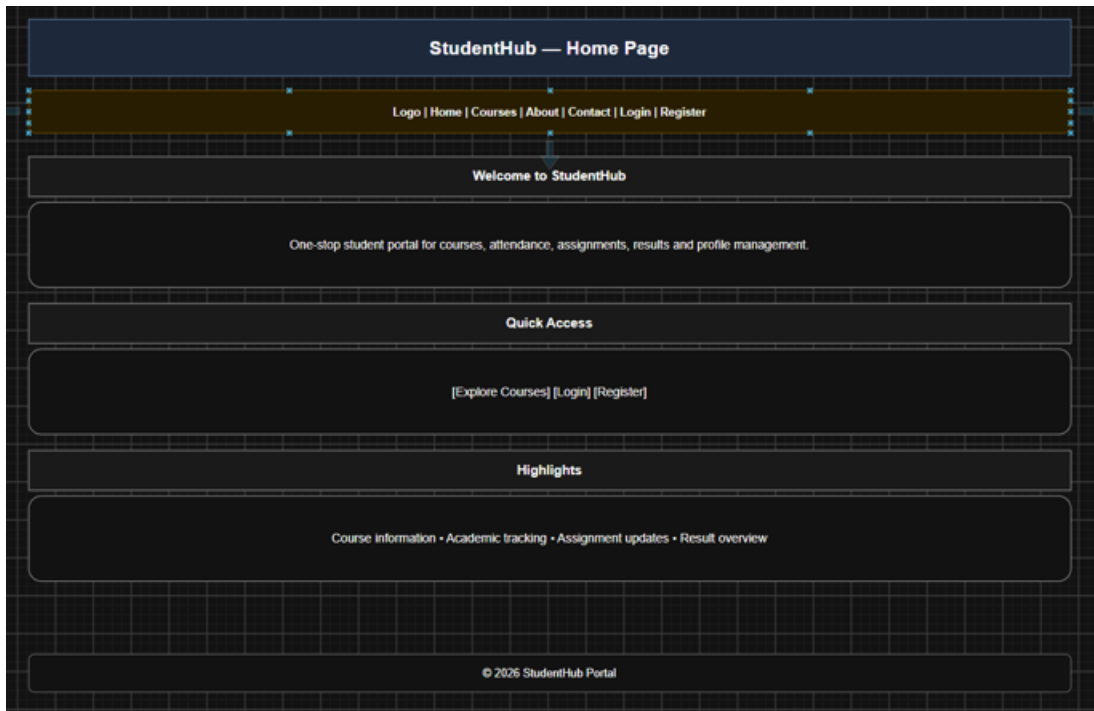
B. Evidence of Work (Screenshots):

Sitemap: [StudentHubPortal/Wireframe & Sitemap/Sitemap.txt at main · aryan150806-wq/StudentHubPortal](#)

A screenshot of a GitHub repository page for the file 'Sitemap.txt'. The repository is named 'StudentHubPortal' and the file is located at 'Wireframe & Sitemap / Sitemap.txt'. The file was created by 'aryan150806-wq' 5 days ago. The file content is a text-based sitemap for a 'Student Hub Portal'. It lists a 'HOME' page and a series of sub-pages under a 'Login' section: 'Register', 'Dashboard', 'Attendance', 'Profile', 'Assignments', 'Courses', 'Results', and 'Contact'. The file is 17 lines long, 15 lines of code, and 233 bytes.

```
1 Sitemap:
2
3 Student Hub Portal
4
5     HOME
6     |
7     |--Login
8     |--Register
9     |--Dashboard
10    | |
11    | |--Attendance
12    | |--Profile
13    | |--Assignments
14    | |--Courses
15    | |--Results
16    |
17    |--Contact
```

Wireframe: [StudentHubPortal/Wireframe & Sitemap/wireframes.docx at main · aryan150806-wq/StudentHubPortal](#)



StudentHub — Register Page

Logo | Home | Login | Register

Create Account

Full Name: _____ Email: _____ Student ID: _____
Password: _____

Academic Details

Program: _____ Semester: _____ Department: _____ Division: _____

Actions

[Create Account] [Back to Login]

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StudentHub — Dashboard Page

Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile | Logout

Welcome, Student

Student Name | ID | Semester | Department

Academic Summary

Attendance: 82% Assignments: 6 Pending CGPA: 8.4

Quick Modules

[Courses] [Attendance] [Assignments] [Results] [Profile]

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StudentHub — Course Page

Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile

My Courses

Course Code | Course Name | Faculty | Credits CS301 | Web Development | Faculty Name | 4 CS302 | DSA | Faculty Name | 4

Course Details

[View Materials] [View Assignments] [View Faculty]

Actions

[Back to Dashboard]

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StudentHub — Attendance Page

Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile

Attendance Overview

Overall Attendance: 82% Present: 41 Absent: 9 Total: 50

Subject-wise Attendance

Subject | Present | Total | Percentage Web Development | 18 | 20 | 90% DSA | 16 | 20 | 80% DBMS | 7 | 10 | 70%

Actions

[Back to Dashboard]

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StudentHub — Assignment Page

Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile

Assignment List

Assignment | Course | Due Date | Status Practical 1 | Web Dev | 10 Aug | Pending Lab 2 | DSA | 12 Aug | Submitted

Assignment Actions

[View Details] [Submit Assignment] [View Submission]

Actions

[Back to Dashboard]

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StudentHub — Result Page

Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile

Semester Result

Semester: 3 SGPA: 8.6 CGPA: 8.4

Subject Results

Subject | Internal | External | Grade Web Development | 42 | 46 | A DSA | 40 | 44 | A

Actions

[Download Result] [Back to Dashboard]

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The image displays two wireframe screenshots of a web portal titled 'StudentHub'.

StudentHub — Profile Page

The top navigation bar includes links: Logo | Dashboard | Courses | Attendance | Assignments | Results | Profile | Logout.

The main content area is divided into sections:

- Student Profile**: Contains input fields for Name, Student ID, Email, and Phone.
- Academic Profile**: Displays the text 'Program: B.Tech Department: Computer Engineering Semester: 3'.
- Actions**: Contains buttons for [Edit Profile], [Change Password], and [Back to Dashboard].

The footer includes the copyright notice: © 2026 StudentHub Portal.

StudentHub — Contact Page

The top navigation bar includes links: Logo | Home | Courses | About | Contact | Login | Register.

The main content area is divided into sections:

- Contact StudentHub**: A header section.
- Contact Form**: Contains input fields for Name, Email, Subject, and Message.
- Actions**: Contains buttons for [Send Message] and [Back to Home].

The footer includes the copyright notice: © 2026 StudentHub Portal.

A 'Snipping Tool' watermark is visible in the bottom right corner of the second screenshot, along with the text 'Screenshot copi' and 'Automatically si'.

C. GitHub Repository Link:

[aryan150806-wq/StudentHubPortal: Web Development Framework P](https://github.com/aryan150806-wq/StudentHubPortal)

D. Challenges Faced:

During the process of creating wireframe, I faced problem creating the website structure in the site draw.io .With repeated tries I was able to manage and use the tool effectively for the creation of wireframe.

E. Learning Outcomes:

I learned what different steps or processes are there present before actually starting to make the project. Such as requirement analysis and how is it done.

I learned about Sitemap and wireframe and its importance before the actual project creation starts.

I also learned about Github and how does it help us to collaborate on a larger project with many team members.

Faculty Remarks: